

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

WEEK 1

Step 1 Database Setup

Query 1: Create the database

```
[mysql> CREATE DATABASE IF NOT EXISTS bookflow_db;  
Query OK, 1 row affected (0.00 sec)
```

Query 2: Select the database

```
[mysql> USE bookflow_db;  
Database changed
```

Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table

```
mysql> CREATE TABLE books (  
->   book_id INT AUTO_INCREMENT PRIMARY KEY,  
->   title VARCHAR(255) NOT NULL,  
->   isbn VARCHAR(13) NOT NULL UNIQUE,  
->   published_year INT,  
->   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

Query 4: Create the members table

```
mysql> CREATE TABLE members (  
->   member_id INT AUTO_INCREMENT PRIMARY KEY,  
->   full_name VARCHAR(100) NOT NULL,  
->   email VARCHAR(150) NOT NULL UNIQUE  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

Query 5 (verification): Inspect the table structures

```
[mysql> DESCRIBE books;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra      |  
+-----+-----+-----+-----+-----+-----+  
| book_id    | int       | NO   | PRI | NULL    | auto_increment |  
| title      | varchar(255) | NO   |     | NULL    |              |  
| isbn       | varchar(13) | NO   | UNI | NULL    |              |  
| published_year | int       | YES  |     | NULL    |              |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.01 sec)
```

```
[mysql> DESCRIBE members;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra      |  
+-----+-----+-----+-----+-----+-----+  
| member_id  | int       | NO   | PRI | NULL    | auto_increment |  
| full_name  | varchar(100) | NO   |     | NULL    |              |  
| email      | varchar(150) | NO   | UNI | NULL    |              |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.01 sec)
```

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Step 3 Initial Seeding (Valid Data)

Query 6: Insert 3 books

```
mysql> INSERT INTO books (title, isbn, published_year) VALUES
-> ('The Alchemist', '9780061122415', 1988),
-> ('Clean Code', '9780132350884', 2008),
[ -> ('Atomic Habits', '9780735211292', 2018);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

Query 7: Insert 3 members

```
mysql> INSERT INTO members (full_name, email) VALUES
-> ('Anil Kumar', 'anil.kumar@example.com'),
-> ('Priya Sharma', 'priya.sharma@example.com'),
[ -> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

5. Step 4 Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

Test 2: NULL title (NOT NULL violation)

Test 3: Future publication year (CHECK violation)

Test 4: Duplicate email (UNIQUE violation)

```
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES (NULL, '9999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES ('Time Traveler', '8888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
[ -> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql> exit;
Bye
```